

25AM206 OBJECT ORIENTED PROGRAMMING

OBJECT-ORIENTED PROGRAMMING

Hours Per Week :

L	T	P	SL	C
2	0	4	2	4

PREREQUISITE KNOWLEDGE: : Basic of C, Classes, Objects and Inheritance.

COURSE DESCRIPTION AND OBJECTIVES:

This course is about the fundamentals of Object-Oriented Programming (OOP) Concept and OOP-based software development methodology. Java as a class-based and pure OOP language is used to demonstrate and implement appropriate concepts and techniques. The students are exposed to the concepts, fundamental syntax, and the thought processes behind object-oriented programming. By end of the course, students will acquire the basic knowledge and skills necessary to implement object-oriented programming techniques in software development using Java.

MODULE-1

UNIT-1

18L+0T+12P+12SL=42 Hours

INTRODUCTION:

History of Java, Byte code, Three OOP principles, Data Types, Variables and Arrays, Operators, control statements, Introduction to Classes and Methods, Method Overloading, Constructors, Constructor overloading, Usage of static, this key word, Garbage collection, String handling and String tokenizer.

Sarva-Tantra-Siddhanta.

UNIT -2

INHERITANCE AND INTERFACES:

Types of inheritance, Usage of This, Static, Final keyword, Method Overriding, Dynamic method dispatch. Abstract classes. Packages-Defining a Package, Access Protection, Importing Packages. Interfaces -Defining, implementing, nesting and applying interfaces.

PRACTICES:

- There is a telecommunication company called "Powered Air" who have approached you to build their Interactive Voice Response (IVR) system. write a Java program and be able to provide the following menu (given below):

Note: User should provide an input for each menu display. Welcome to Powered Air service.

What would you like to do?

- | | | |
|---|--------------------------|--------------------|
| a. Know my balance. | b. Know my validity date | |
| c. Know number of free calls available. | d. More | |
| 1. Prepaid Bill Request | 2. Customer Preferences | 3. GPRS activation |
| 4. Special Message Offers | 5. Special GPRS Offers | 6. 3G Activation |

Go back to Previous menu.

You are free to display your own messages in this IVR.

- Create a class Rectangle. The class has attributes length and width. It should have methods that calculate the perimeter and area of the rectangle. It should have read Attributes method to read length and width from user.
- Create a class called Customer with data members account_number, balance (initialize with 10000), and member functions print(), deposit(), and withdraw(). Print method display account number and balance. If withdraw amount is less than current balance while withdrawing, throw an exception "In Sufficient Funds". If the input is 1 do print. If the input is 2 withdraw (). If the input is 3 deposit. If the input is 4 terminate program.

SKILLS:

- ✓ Applying OOP principles to design efficient solutions.
- ✓ Understanding best practices in coding, debugging, and testing.
- ✓ Familiarity with collections framework, AWT and swings.

- Implement a Java program which accepts age as input from the user and throws an exception "Not Eligible to Vote" when age is ≤ 18 otherwise print "Eligible to Vote".
- Using inheritance, one class can acquire the properties of others. Consider a class Animal that has only one method "walk". Next, create a Bird class that also has a fly method. Finally, create a bird object that can both fly and walk
- Using inheritance, Write the following code in your editor : 1. A class named Arithmetic with a method named "add" that takes integers as parameters and returns an integer denoting their sum. 2. A class named Adder that inherits from a superclass named Arithmetic. Note: Your classes should not be Public.
- When a subclass inherits from a superclass, it also inherits its methods; however, it can also override the superclass methods (as well as declare and implement new ones). Consider the Sports class having methods getName()[which returns name of sport] and getNumberOfTeamMembers()[which returns no of team members] create a Soccer class that inherits from the Sports class. We can override the get Name method and return a different subclass-specific string and override getNumberOfTeamMembers method and return no of team members
- You are given an interface Advanced Arithmetic which contains a method signature `int divisor_sum(int n)`. You need to write a class called My Calculator which implements the interface. divisor_sum function just takes an integer as input and return the sum of all its divisors. For example divisors of 6 are 1, 2, 3 and 6, so divisor_sum should return 12. The value of n will be at most 1000.
- Implement a Java program for the following a) Creation of simple package. b) Accessing a package.
- Implement a Java program to read two numbers a,b from user and perform division a/b, if the user passes b value as zero, handle the exception using try and catch otherwise display the result.
- Create a class called Customer with data members account_number, balance (initialize with 10000), and member functions print(), deposit(), and withdraw(). Print method display account number and balance. If withdraw amount is less than current balance while withdrawing, throw an exception "Insufficient Funds". If the input is 1 do print. If the input is 2 withdraw (). If the input is 3 deposit. If the input is 4 terminate program

MODULE-2**UNIT-1****27L+0T+18P+27SL = 72 Hours****EXCEPTION HANDLING AND MULTI THREADING:**

Exception Handling fundamentals, Exception types, Usage of try, catch, throw, throws and finally keywords, Java's Built-in Exceptions, Creating your own Exceptions Multithreading-Process and Thread, Thread life cycle, creating multiple threads using Thread class and Runnable interface and Synchronization, Inter thread communication.

UNIT-2**EVENT HANDLING AND AWT CONTROLS:**

Delegation event model -Events, Event sources, Event Listeners, Event classes, handling Mouse Events and Keyboard Events. Awt controls: Windows Fundamentals, Label, Button, Check Box, Choice, List Scroll Bars, Text Field, Text Area, Layout Managers.

UNIT-3**THE COLLECTION FRAMEWORK:**

List – Array List, Linked List, Vector, Set - HashSet, TreeSet, Map - HashTable, HashMap, accessing a collection via an Iterator, comparator, comparable. Exploring swings - JLabel, ImageIcon, Swings buttons, JTabbedPane, JScrollPane, JList, JComboBox.

COLLECTION INTERFACES AND CLASSES:

List – Array List, Linked List, Vector, Set - HashSet, TreeSet, Map - HashTable, HashMap. Exploring Swings : JLabel, ImageIcon, Swings buttons, JTabbedPane, JScrollPane, JList, JComboBox.

PRACTICES:

- Implement a Java program for handling mouse events when the mouse entered, exited, clicked,

pressed, released, dragged and moved in the client area.

- Implement a Java program for handling key events when the key board is pressed, released, typed.
- Implement a Java swing program that reads two numbers from two separate text fields and display sum of two numbers in third text field when button “add” is pressed.
- Implement a Java program to design student registration form using Swing Controls. The form which having the following fields and button “save”. Form Fields are: Name, RNO, Mail id, Gender, Branch, and Address.
- Implement a java program using swings to design a multiple choice question having three options (use radio button), display the message using dialog box “Your answer is wrong” if the user selects wrong option otherwise display, “Your answer is correct.”

COURSEOUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply classes, methods, constructors and string Tokenizer for efficient text manipulations in Java	Apply	1,2
2	Implement event driven programming using Java's delegation event model for handling user interactions.	Apply	1,2
3	Utilize access control mechanisms to develop secure and scalable real-time applications.	Apply	1,2,4
4	Design and develop Java applications to solve real world problems by using modern tools and collection framework	Create	1,3,5
5	Develop GUI based applications using swings for web and desktop environments.	Create	1,3,5

MAPPING OF SDGS AND IKS

Sustainable Development Goals (SDGs)	Indian Knowledge System
SDG 4 (Quality Education)	Yes
SDG 9 (Industry, Innovation, and Infrastructure)	
SDG 11 (Sustainable Cities and Communities)	

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXT BOOKS:

1. Herbert Schildt, “Java: The Complete Reference”, 13th Edition, McGraw Hill, Education, January 2024.

REFERENCE BOOKS:

1. Balagurusamy, “Programming with Java”, 7th Edition, McGraw Hill, 2023.
2. M.T. Somashekara, D. S. Guru, K.S. Manjunatha, “Object-Oriented Programming with Java”, 1st Edition, PHI Learning, 2017.
3. Mark Lassoﬀ, “Java Programming for Beginners: Learn the fundamentals of programming with Java”, 1st Edition, Packt Publishing Limited, 2017.